MINISTRY OF EDUCATION AND SCIENCE OF THE RUSSIAN FEDERATION

Federal State Autonomous Educational
Institution of Higher Education "Southern Federal University"
(SOUTHERN FEDERAL UNIVERSITY)

COURSE SYLLABUS FOR

Operating Systems (Операционные системы)

Подписано электронной подписью:

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I. GOALS AND OBJECTIVES OF MASTERING THE DISCIPLINE

Objectives of mastering the discipline:

– obtaining by students theoretical knowledge of the principles of organization, the basics of construction, features of the functioning and use of modern fixed assets;

– acquisition of practical skills and abilities in installing, configuring, configuring, protecting, maintaining and using the OS in various modes of operation.

Objectives of mastering the discipline:

– to familiarize students with the purpose, functions, types, classification, principles of construction and modes of operation of the OS;

– to give an overview of modern operating systems, to familiarize students with the specifics, differences in the properties and capabilities of popular operating systems; trends in the development of fixed assets at the present stage;

– to teach students to understand and take into account the conceptual foundations of the OS – resource, process, functional components of the OS; Interrupt and virtualization concepts disciplines of resource allocation; means of interaction between the user and the system; requirements for modern fixed assets and trends in their development;

– to teach students to understand the architectures of modern operating systems, features of compatibility, hardware dependency and portability of the OS, directions and means of virtualization, the capabilities of virtual machines and hypervisors;

– to teach students to understand and use the internal mechanisms of the OS – process and flow management tools (features and algorithms of scheduling and dispatching), interrupt dispatching, synchronization tools, as well as tools for managing memory, I/O devices and files, file systems and security tools;

– to instill in students the skills and practical skills of qualified work in the environment of various operating systems, effective organization of calculation and input-output processes in solving professional problems; optimal use of computing resources, file systems and security tools in the OS;

– to instill in students the skills and abilities to use artificial intelligence systems to analyze ways to solve problems of effective use of the capabilities of operating systems.

II. THE PLACE OF THE DISCIPLINE IN THE STRUCTURE OF THE EDUCATIONAL PROGRAM

The discipline belongs to the module of university-wide disciplines.

To study this discipline, it is necessary to have knowledge, skills and abilities formed by the previous elements of the educational program:

Name disciplines (modules), practices	Required knowledge, skills, and abilities
Discrete Mathematics	Knowledge: – Fundamentals, Apparatus and Methods of Discrete Mathematics
	Skills: – apply discrete mathematical models in solving various problems
	Skills: – analysis of discrete structures and models
Algorithmization and Programming	Knowledge: – the basics of algorithmization and programming, methods of construction and structures of algorithms and programs, tools and programming systems, the basics of C++ and Python programming languages
	Skills: – use tool software to create programs in C++ and Python
	Skills: – algorithmization of various problems, development and debugging of programs for their solution.
	Knowledge: – principles of functioning of generative artificial intelligence systems

Name disciplines (modules), practices	Required knowledge, skills, and abilities
Generative Artificial Intelligence and Machine Learning	Skills: – use generative artificial intelligence systems to solve typical problems of setting up and using the OS
	Skills: – solving various problems using AI systems

Knowledge, skills and abilities formed by this discipline will be required when mastering the following elements of the educational program:

– disciplines "Computer Networks", "Databases and DBMS", "Technologies for Storing and Processing Big Data", "Tools and Engineering Practices for the Development of AI Systems", "Implementation, Maintenance, Configuration and Operation of Information Systems";

– creative project "Development of Elements of Support for Multimedia and Remote Technologies";

– all types of practices;

– GIA.

III. REQUIREMENTS FOR THE RESULTS OF MASTERING THE DISCIPLINE

The development of the discipline is aimed at the formation of the following competencies in accordance with the educational standard and educational program:

List of planned learning outcomes in the discipline correlated with indicators of competence achievement

Competence	Indicators of Competency Achievement	Learning outcomes
CPC-2. Able to understand the principles of modern information technology and software, including domestic production, and use them in solving the problems of professional activity	CPC-2.2. Applies modern information and communication technologies, hardware and software in solving the problems of professional activity	Knowledge: – the possibilities and features of the use of modern and promising information and communication technologies in solving the problems of professional activity in the environment of various operating systems; – features of the use of operating systems of various classes and architectures in solving applied problems; – purpose, capabilities and features of the use of hardware and software, including domestic production, to support and maintain the OS.
		Skills: – to apply modern and promising information and communication technologies laid down in the OS by their developers; – use the appropriate necessary hardware and software of the environment of the operating systems used.
		Skills: – effective use of software and hardware complexes of various operating systems in solving problems of professional activity.
CPC-5. Able to install software and hardware for information and automation systems	CPC-5.1. Installs software and hardware based on modern standards of information interaction of systems	Knowledge: – features and capabilities of administering the OS environment.
		Skills: – to administer the software and information structures of the OS on the basis of modern standards of information interaction of systems.
		Skills: – administration of the MS Windows operating system environment.

Competence	Indicators of Competency Achievement	Learning outcomes
CPC-7. Able to participate in setting up and adjustment of software and Hardware Complexes	CPC-7.1. Performs parametric configuration of software for information and automated systems	Knowledge: – capabilities and tools for parametric configuration of the OS and software tools to support the environment of its operation.
		Skills: – perform parametric OS configuration; – perform parametric configuration of software to support the functioning of the OS environment.
		Skills: – effective parametric configuration of the operating environment of the MS Windows operating system.
PC-9. PL-1 Capable of using the Python programming language to solve problems in the field of AI	ПК-9.4. PL-1.4 Designs distributed computing systems in Python to efficiently handle a large number of tasks	Knowledge: – principles of operation of distributed computing systems.
		Skills: – design, develop and debug complex software systems.
		Skills: – development of Python programs for managing the execution of computational work.
PC-10. PL-3 Able to use C/C++ programming languages to solve problems in the field of AI	ПК-10.1. PL-3.1 Develops and debugs effective multithreaded C++ solutions, tests, tests and evaluates the quality of such solutions	Knowledge: – the main problems of designing and implementing multithreaded computing and methods for solving them.
		Skills: – Develop efficient multithreaded solutions in C++.
		Skills: – \u2012 debugging and testing of multithreaded solutions, assessing the quality of such solutions.
	ПК-10.2. PL-3.2 Develops and debugs C++ AI systems for specific hardware platforms	Knowledge: – The characteristics and capabilities of the hardware platforms used to implement AI systems.

Competence	Indicators of Competency Achievement	Learning outcomes
	with processing power limitations, including embedded systems	Skills: – Develop C++ programs that make efficient use of limited hardware.
		Skills: – Optimization of developed programs in terms of memory and execution time.
	ПК-10.3. PL-3.3 Develops and debugs C++ solutions that use GPUs and FPGAs to massively parallelize computations within a common AI system, using both off-the-shelf solutions and developing their own	Knowledge: – features of GPU and FPGA programming.
		Skills: – Develop C++ programs that use GPUs and FPGAs.
		Skills: – use of ready-made solutions within the framework of a common AI system.
PC-14. LLM-5 Organizes interaction with generative models through the design, analysis and application of prompts	PC-14.2. LLM-5.2 Embeds Prompts in the Interaction Pipeline	Knowledge: – features of different AI systems and the optimal structure of prompts for their effective use.
		Skills: – organize interaction with generative AI models, including the use of query pipelineization.
		Skills: – compiling prompts for AI systems.
	SC-14.3. LLM-5.3 Configures API for working with LLM	Knowledge: – API capabilities for specific LLMs.
		Skills: \u2012 configure the API to work with LLM.
		Skills: \u2012 use APIs when developing solutions related to LLM.

IV. CONTENT AND STRUCTURE OF THE DISCIPLINE

The complexity of the discipline is 5 credits, 180 hours,
including 1 credit, 36 hours for the exam.

Form of intermediate certification: exam .

4.1. The content of the discipline, structured by topics (educational meetings) for _3_ semester of full-time education

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Practical	Cases	Consulting					
Module 1. Operating System Basics										
1	**Basic information about fixed assets.** Purpose and definitions of fixed assets.	2	–	–	–	–	Informational lecture	–	–	–
2	**Case No 1. Selection of dispatching algorithms.** Receiving an assignment, discussing a case.	–	–	2	–	4	Case	–	–	–
3	**Basic information about fixed assets.** The main functions of the OS. Units of computational work. Modes provided by the OS	2	–	–	–	2	Informational lecture	–	–	–
4	**Case No 1. Selection of dispatching algorithms.** Execution of the case. Download results.	–	–	2	–	2	Case	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Practical	Cases	Consulting					
5	**Basic information about fixed assets.** Classification of fixed assets. Overview of modern operating systems.	2	–	–	–	2	Informational lecture	–	–	–
6	**Case No 1. Selection of dispatching algorithms.** Selective protection of case reports.	–	–	2	–	2	Case	Case No 1	Preparation and submission of a case report	6
7	**Basic information about fixed assets.** Requirements for fixed assets. General trends in the development of modern operating systems	2	–	–	–	2	Informational lecture	–	–	–
8	**Case No 2. Allocate memory to processes.** Receiving an assignment, discussing a case.	–	–	2	–	2	Case	–	–	–
9	**Conceptual foundations of the OS.** Resources and processes, their types and features. Relations between interrelated processes. Flows.	2	–	–	–	2	Informational lecture	–	–	–
10	**Case No 2. Allocate memory to processes.** Execution of the case. Download results.	–	–	2	–	2	Case	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Practical	Cases	Consulting					
11	**Conceptual foundations of the OS.** Functional components of the OS. OS subsystems. API. Graphical user interfaces (GUIs).	2	–	–	–	2	Informational lecture	–	–	–
12	**Case No 2. Allocate memory to processes.** Selective protection of case reports.	–	–	2	–	2	Case	Case No 2	Preparation and submission of a case report	7
13	**Conceptual foundations of the OS.** The concept of interruption. The concept of virtualization. Disciplines of resource allocation. Means of user interaction with the OS.	2	–	–	–	2	Informational lecture	–	–	–
14	**Case No 3. Data protection**. Receiving an assignment, discussing a case.	–	–	2	–	2	Case	–	–	–
15	**Conceptual foundations of the OS.** Basics of security in the OS. Security tasks. Threats.	2	–	–	–	2	Informational lecture	–	–	–
16	**Case No 3. Data protection**. Execution of the case. Download results.	–	–	2	–	2	Case			

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Practical	Cases	Consulting					
17	**Conceptual foundations of the OS.** Access control models. Hidden channels. Test on the basics of operating systems	2	–	–	–	2	Informational lecture	Test No 1	Written response	10
18	**Case No 3. Data protection.** Selective protection of case reports.	–	–	2	–	2	Case	Case No 3	Preparation and submission of a case report	7
Module 2. Operating System Architecture and Mechanisms										
19	**OS architecture.** Monolithic OS kernel and privileges, multi-layer OS structure. Hardware dependency and OS portability.	2	–	–	–	2	Informational lecture			
20	**Case No 4. Elimination of the effect of racing (mutual exclusion).** Receiving an assignment, discussing a case.	–	–	2	–	2	Case			
21	**OS architecture.** Microkernel-based architecture. New kernel and OS architecture options. Compatibility and multiple PPPs. Directions of virtualization. Hypervisors	2	–	–	–	2	Informational lecture			

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Practical	Cases	Consulting					
22	**Case No 4. Elimination of the effect of racing (mutual exclusion).** Execution of the case. Download results.	–	–	2	–	2	Case			
23	**Process and flow management.** Plan and dispatch processes and flows. Dispatching and accounting for interrupt priorities in the OS.	2	–	–	–	2	Informational lecture			
24	**Case No 4. Elimination of the effect of racing (mutual exclusion).** Selective protection of case reports.	–	–	2	–	2	Case	Case No 4	Preparation and submission of a case report	6
25	**Process and flow management.** System call dispatching. Synchronization of processes and threads.	2	–	–	–	2	Informational lecture			
26	**Case No 5. Organization of process interaction.** Receiving an assignment, discussing a case.	–	–	2	–	2	Case			
27	**Process and flow management.** Races, semaphores, synchronizing OS objects, signals.	2	–	–	–	2	Informational lecture			

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Practical	Cases	Consulting					
28	**Case No 5. Organization of process interaction.** Execution of the case. Download results.	–	–	2	–	2	Case			
29	**Memory management.** Types of addresses, structuring the virtual address space. Memory allocation algorithms (RP). Swapping and Virtual Memory	2	–	–	–	2	Informational lecture			
30	**Case No 5. Organization of process interaction.** Selective protection of case reports.	–	–	2	–	2	Case	Case No 5	Preparation and submission of a case report	7
31	**Memory management.** Page RP. Segmental RP. Segment-page RP. Shared memory segments.	2	–	–	–	2	Informational lecture			
32	**Case No 6. File system structure.** Receiving an assignment, discussing a case.	–	–	2	–	2	Case			
33	**Input/Output (OCC) control.** Tasks of the CB for the management of hydrocarbons. A multi-layer model of the I/O subsystem.	2	–	–	–	2	Informational lecture			

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Practical	Cases	Consulting					
34	**Case No 6. File system structure.** Execution of the case. Download results.	–	–	2	–	2	Case			
35	**File management.** File systems (FS). Logical and physical FS organization. Test work on the architecture and mechanisms of the OS	2	–	–	–	2	Informational lecture	Test No 2	Written response	10
36	**Case No 6. File system structure.** Selective protection of case reports.	–	–	2	–	2	Case	Case No 6	Preparation and submission of a case report	7
Intermediate certification		–	–	–	–	36		Exam Questions and Tickets for Semester 3	*Exam*	40
Total Hours		**36**		**36**		**108**		**–**		**100**

4.2. Plan of extracurricular independent work

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
Module 1. Operating System Basics						
1	OS Basics	3	1–4	– Study and revision of lecture material	6	[3]-[6]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
2	Selection of dispatching algorithms	3	1–3	– preparation for the implementation of the case, preparation of a report on the implementation of the case, preparation for the defense of the report on the implementation of the case.	6	[3]-[10]
3	Allocate memory to processes	3	4–6	– preparation for the implementation of the case, preparation of a report on the implementation of the case, preparation for the defense of the report on the implementation of the case.	6	[3]-[10]
4	Data protection	3	5–9	– preparation for the implementation of the case, preparation of a report on the implementation of the case, preparation for the defense of the report on the implementation of the case.	8	[3]-[10]
5	Conceptual foundations of the OS.	3	7–9	– study and repetition of the material of lectures.	10	[8]
Module 2. Operating System Architecture and Mechanisms						
6	OS architecture	3	10–11	– Study and revision of lecture material	2	[1]-[2], [3]-[6]
7	Mutual exclusion	3	10–12	– preparation for the implementation of the case, preparation of a report on the implementation of the case, preparation for the defense of the report on the implementation of the case.	6	[3]-[10]
8	Process and flow management	3	12–14	– Study and revision of lecture material	4	[1]-[2], [3]-[6]
9	Organization of process interaction.	3	13–15	– preparation for the implementation of the case, preparation of a report on the implementation of the case, preparation for the defense of the report on the implementation of the case.	6	[3]-[10]
10	Memory Management	3	15–16	– Study and revision of lecture material	2	[1]-[2], [3]-[6]
11	File system structure.	3	16–18	– preparation for the implementation of the case, preparation of a report on the implementation of the case, preparation for the defense of the report on the implementation of the case.	6	[3]-[10]
12	I/O Device Management	3	17	– study and repetition of the material of lectures.	5	[1]-[2], [3]-[4]
13	File Management	3	18	– study and repetition of the material of lectures; – Preparation for the test	5	[1]-[2], [3]-[6]
Exam preparation					**36**	**[1]–[8]**
The total labor intensity of independent work in the discipline					**108**	**–**

V. EDUCATIONAL AND METHODOLOGICAL SUPPORT OF THE DISCIPLINE

5.1. References

1. Nuzhnov E.V., Danilchenko V.I. Operating systems [Elektronnyi resurs] : uchebnoe posobie : v 2 ch. **Part 1**. Fundamentals of Operating Systems / E. V. Nuzhnov, V. I. Danilchenko; Southern Federal University. – Rostov-on-Don; Taganrog: Publishing House of the Southern Federal University, **2023**. – 145 p. - http://hub.lib.sfedu.ru/repository/material/801312574/.
2. Nuzhnov E.V. Operating systems [Elektronnyi resurs]. Uchebnoe posobie. Part 2. Architecture and mechanisms of operating systems. – Rostov-on-Don: SFedU Publishing House, 2014. – 163 p. – Mode of access: URL http://ntb.tti.sfedu.ru/UML/UML_5418_2.pdf.

5.2. Further reading

3. Nuzhnov E.V. Operating Systems [Elektronnyi resurs]: Uchebno-metodicheskoe posobie po organizatsii i ispolnenii samostoyatel'noy raboty studentov [Operating systems]: Educational and methodological manual for the organization and performance of students' independent work. – Taganrog: TTI SFU Publ., 2010. – 52 p. – Rezhim dostupa: URL http://ntb.tti.sfedu.ru/UML/UML_3127_3.pdf.
4. Nazarov S.V. Sovremennye operativnye sistemy [Elektronnyi resurs] / S.V. Nazarov; A.I. Shirokov - Moscow: Internet University of Information Technologies, 2011. – 280 p. – Mode of access: URL http://biblioclub.ru/index.php?page=book&id=233197.
5. Safonov V.O. Osnovy sovremennykh operatsionnykh sistem [Elektronnyi resurs] / V.O. Safonov - Moscow: Internet-universitet informatsionnykh tekhnologii, 2011. – 584 p. – Rezhim dostupa: URL http://biblioclub.ru/index.php?page=book&id=233210.
6. Gritsenko - Tomsk: Tomsk State University of Control Systems and Radioelectronics, 2009. - 235 p. - Available at: URL http://biblioclub.ru/index.php?page=book&id=208655.
7. ChatGPT – Available at: https://chatgpt.com/ (accessed: 07.11.2025).
8. GigaChat – Russian-language neural network from Sberbank – Available at: URL: https://giga.chat/ (accessed: 07.11.2025).
9. DeepSeek – V nezvestnost' – Rezhim dostupa: URL:https://chat.deepseek.com/ (accessed 07.11.2025).
10. Qwen Chat – Available at: https://chat.qwen.ai/ (accessed: 07.11.2025).

5.3. Periodicals

- Journal "Programming". – https://www.ispras.ru/programming/;
- Mir PK Magazine. – http://www.osp.ru/pcworld/;
- The journal "Open Systems. DBMS". – http://www.osp.ru/os/;
- Computerworld. Russia" – http://www.osp.ru/cw/#/home;
- Computerra Magazine. – http://www.computerra.ru/;
- ComputerPress Magazine. – http://www.compress.ru/;
- Byte Magazine. – http://www.bytemag.ru/;
- Windows IT Pro Magazine. – http://www.osp.ru/win2000/#/home;
- Digital World Magazine. – http://www.dgl.ru/;
- InZone Magazine. – http://www.andrakov.narod.ru/.

5.4. List of Internet resources

Text resources:

\u2012 Safonov V. Osnovy sovremennykh operatsionnykh sistem [Fundamentals of Modern Operating Systems], 2010. –http://www.intuit.ru/department/os/bmos/; http://www.intuit.ru/studies/courses/641/497/info.

\u2012 Nazarov S.V., Shirokov A.I. Modern Operating Systems, 2010. – http://www.intuit.ru/department/os/modernos/; http://www.intuit.ru/studies/courses/631/487/info.

\u2012 Popov A. Command Line and Windows Scripts. – http://www.intuit.ru/studies/courses/1059/225/info.
\u2012 Kostromin V. Fundamentals of Work in OS Linux. – http://www.intuit.ru/studies/courses/91/91/info.
\u2012 Kuryachiy G., Maslinsky K. Operating system Linux. – http://www.intuit.ru/studies/courses/37/37/info.

Video resources:
\u2012 Karpov V. Fundamentals of Operating Systems. – http://www.intuit.ru/studies/courses/1088/322/info.
\u2012 Nazarov S.V. Operating Environments, Systems and Shells. – http://www.intuit.ru/studies/courses/492/348/info.

VI. MATERIAL AND TECHNICAL SUPPORT OF THE DISCIPLINE

The following premises, equipment and software are used in the implementation of the discipline:

No p/n	Type of classroom lesson	Recommended audience	
		Type	Equipment
1	Lectures	lecture-type auditoriums (G-309, G-333, L-230, etc.)	– interactive whiteboard or projector - 1 pc., – teacher's computer - 1 pc.; – Microsoft Windows, Microsoft Office – PowerPoint
3	Cases	specialized classrooms (laboratories, computer classes), interdepartmental and departments of VT, MOP COMPUTERS, SAiT, CAD	\u2012 interactive whiteboard or projector - 1 pc., teacher's computer - 1 pc.; \u2012 Microsoft Windows distributions (free license under the Microsoft Imagine program); \u2012 Acronis True Image (free versions and licenses); \u2012 Linux distributions (free license, open source); \u2012 Oracle Solaris distributions (free license); \u2012 free virtual machines (Virtual Box, VMware Workstation Player); \u2012 free operating systems and applications: Knoppix Live CD, Simply Linux, Maniaro).
4	Independent work		

VII. METHODICAL INSTRUCTIONS FOR STUDENTS ON MASTERING THE DISCIPLINE

The discipline "Operating Systems" is studied in the 3rd semester and includes two modules: Module 1 "Fundamentals of Operating Systems", Module 2 "Architecture and Mechanisms of Operating Systems".

The educational process in the discipline includes classroom classes (lectures, cases) and independent work. Intermediate certification in the discipline is an exam. Teachers control the attendance of all types of classroom classes.

The system of university education is based on a rational combination of several types of classroom training and independent extracurricular and classroom work, each of which has a certain specificity.

Preparation for lectures and the student's work at the lecture.

The use of advanced computer and information technologies has opened up new opportunities in the student's work. Before the first lecture, each student can download PDF files of lecturers' textbooks for each module of the discipline from the SFedU electronic library, and then preliminarily work through the educational materials before each lecture. Each student is recommended to bring any means (laptop, tablet, smartphone, etc.) to lectures for prompt viewing of these educational materials against the background of the lecturer's story. The lecturer conducts a lecture with the synchronous use of an electronic presentation in the form of slides on a large screen. These two sources of information eliminate the need for students to take notes of the lecture. But each student (if desired) can make short, important and prompt electronic or handwritten notes for himself.

Textbooks and presentations of the lecturer on the modules of the discipline have the following distinctive features: a system of abbreviations (abbreviations), highlighting important information (in bold, underlining, etc.). In addition, the textbook presents a glossary of the main terms of the subject area and tests for the module, and the presentation uses accentuating highlighting in color and signs in the text and illustrations.

Working with the mentioned materials before lectures, the student should use both the main and additional literature recommended by the lecturer. It is such serious, painstaking work with lecture materials that will allow you to deeply master the theoretical material of the discipline.

Preparation for the implementation of cases.

Cases are one of the innovative forms of organizing the educational process in technical specialties, aimed at the creative application of knowledge gained in the process of studying an academic discipline, and obtaining practical skills in analyzing real production problems and solving them using software.

Preparation for solving cases is carried out according to methodological instructions, recommendations of the teacher, as well as basic and additional literature.

At the end of each case, the student is obliged to submit a report to the teacher for verification with subsequent defense.

After the report is drawn up and submitted, the teacher evaluates the student's work by checking the report and defending it (interview).

Tests on the modules of the discipline.

After the completion of the lectures of each module, a written test is carried out in the classroom, the results of which are evaluated by the lecturer. The guide in preparing for the test is the control questions on the module given in the fund of assessment tools.

Preparation for intermediate certification.

The discipline provides for a type of intermediate certification - an exam. Students who score 38-60 points during the semester on the current and midterm control are considered admitted to the exam. Students who, for a good reason, could not score the required number of points, may, in agreement with the teacher, get points before the start of the intermediate certification (exam). The main reference point in preparing for the exam is the questions for the exam given in the fund of assessment tools. Studying the material related to a specific issue, the student should carefully read the recommended literature, identify and consider various approaches to its presentation, analyze the main points, features, advantages and disadvantages. When preparing for the exam ticket, it is recommended to analyze the essence of each question, understand what exactly needs to be presented in the answer, draw up a plan for answering questions and give examples of the use of the theoretical provisions under consideration in practice. It is especially important to show the examiner a confident command of the topic of each question, which is achieved by a qualitative assessment of the state of the question, giving the necessary descriptions and illustrations, examples, attracting successful information from related topics, mentioning developments, details, names, chronology, cause-and-effect relationships with related questions, etc.

VIII. FUND OF ASSESSMENT TOOLS

8.1. Passport of the Appraisal Tools Fund

Nop/n	Competency Achievement Indicator	Name of the Valuation Tool
1	CPC-2.2. Applies modern information and communication technologies, hardware and software in solving the problems of professional activity	– Test No 1; – exam questions and tickets.
2	CPC-5.1. Installs and administers the software and Hardware based on modern standards information interaction of systems	– Test No 2; – Exam questions and tickets
3	CPC-7.1. Performs parametric configuration of software for information and automated systems	– Test No 2; – exam questions and tickets.
4	ПК-9.4. PL-1.4 Designs distributed computing systems in Python to efficiently handle a large number of tasks	– Cases No 1–6 (execution, preparation of the report, defense of the report)
5	ПК-10.1. PL-3.1 Develops and debugs effective multithreaded C++ solutions, tests, tests and evaluates the quality of such solutions	– Cases No 1–6 (execution, preparation of the report, defense of the report)
6	ПК-10.2. PL-3.2 Develops and debugs C++ AI systems for specific hardware platforms with processing power limitations, including embedded systems	– Cases No 1–6 (execution, preparation of the report, defense of the report)
7	ПК-10.3. PL-3.3 Develops and debugs C++ solutions that use GPUs and FPGAs to massively parallelize computations within a common AI system, using both off-the-shelf solutions and developing their own	– Cases No 1–6 (execution, preparation of the report, defense of the report)
8	PC-14.2. LLM-5.2 Embeds Prompts in the Interaction Pipeline	– Cases No 1–6 (execution, preparation of the report, defense of the report)
9	SC-14.3. LLM-5.3 Configures API for working with LLM	– Cases No 1–6 (execution, preparation of the report, defense of the report)

8.2. Case No 1. Selection of dispatching algorithms.

Assessment Tool Type: Case Report

Control type: ongoing

List of controlled topics (modules): Module 1. Operating System Basics

Form of study: full-time

8.3. Case No 2. Allocate memory to processes.

Assessment Tool Type: Case Report

Control type: ongoing

List of controlled topics (modules): Module 1. Operating System Basics

Form of study: full-time

8.4. Case No 3. Data protection.

Assessment Tool Type: Case Report

Control type: ongoing

List of controlled topics (modules): Module 1. Operating System Basics

Form of study: full-time

8.5. Case No 4. Elimination of the effect of racing (mutual exclusion).

Assessment Tool Type: Case Report

Control type: ongoing

List of controlled topics (modules): Module 2. Operating System Architecture and Mechanisms

Form of study: full-time

8.6. Case No 5. Organization of process interaction.

Assessment Tool Type: Case Report

Control type: ongoing

List of controlled topics (modules): Module 2. Operating System Architecture and Mechanisms

Form of study: full-time

8.7. Case No 6. File system structure.

Assessment Tool Type: Case Report

Control type: ongoing

List of controlled topics (modules): Module 2. Operating System Architecture and Mechanisms

Form of study: full-time

8.8. Test No 1 (written answer)

Type of assessment tool: test work

Type of control: milestone

List of controlled topics (modules): Module 1. Operating System Basics

Form of study: full-time

Questions for Test No 1:

1. List the factors that influence the development of the OS.
2. List the means and mechanisms that were milestones in the evolution of the OS.
3. Analyze and compare different definitions of fixed assets.
4. What are the different units of computing work needed for in an OS environment?
5. What units of computing work can be used in an OS environment?
6. What is the difference between a session and a process?
7. How is a task different from a process?
8. What is the difference between flow and process?
9. What criteria for the effectiveness of the Armed Forces emphasize the role of the OS?
10. Compare the different modes of operation of computers provided by the OS.
11. What is multiprogramming and what opportunities does it open?
12. List the features, advantages and disadvantages of multi-program batch mode.
13. What is a multi-program mix and how does its composition affect the overall time to solve problems?
14. Diagram the operation of 3 programs (A: 2 – 9 – 1, B: 1 – 12 – 3, C: 2 – 11 – 2) and CPU load for multiprogram batch mode.
15. List the features, advantages, and disadvantages of real-time mode.

16. What is the criterion for the effectiveness of an aircraft whose operating system operates in real time?
17. Plot the operation of 3 programs (A: 2 – 9 – 1, B: 1 – 12 – 3, C: 4 – 11 – 2) and the CPU load for the time-sharing mode (WFD) at $\Delta tm = 3$ and $\Delta tm = 4$.
18. In which mode (multiprogram, packet, or time splitting) is the processor throughput generally higher and why?
19. What is the criterion for the efficiency of an aircraft whose operating system operates in time-sharing mode?
20. What is the essence of multitasking and what options are known for its implementation?
21. What are the basic features of the OS and why?
22. Prove the importance of all the basic features of the OS.
23. Give a classification of the OS and position the systems you know in it.
24. Explain the main features of LiveCD/DVD OS and Web-OS.
25. Show the importance of considering the principles of building an OS.
26. Explain the essence of the principles of OS construction: frequency, modularity, functional selectivity.
27. Explain the essence of the principles of OS construction: generation, functional redundancy, protection, silence.
28. Explain the essence of the principles of building an operating system: portability, independence of programs from UVS, openness, and scalability.
29. What do the requirements for the OS mean: reliability and fault tolerance, security, extensibility.
30. What the requirements for the OS mean: performance, portability, compatibility.
31. Describe significant events in the evolution of popular operating systems and their families.
32. Describe the process of OS evolution: the development of means to support various modes of their functioning (with examples).
33. Explain the meaning of the trends in the development of fixed assets at the present stage.
34. Characterize and define the resource.
35. What resource is considered: physical, passive, permanent, secondary, simple?
36. Which resource counts: virtual, active, temporary, primary, composite?
37. What is the difference between a reproducible resource and a consumable one?
38. What is the difference between sequentially used and concurrent resources?
39. Show the key meaning of the concepts of "resource" and "process".
40. Demonstrate the role of resource classification features using the example of a substation, printer and other components of the aircraft.
41. Describe the concept and define the process.
42. What states of the process are known and what does the process existence graph show?
43. How does the lifetime interval of a process differ from its trace?
44. Provide a classification of batch processes.
45. What is the difference between equivalent, identical and equal processes?
46. What is the difference between sequential, parallel and combined processes?
47. What processes are distinguished by connectivity?
48. Explain the meaning of the different types of relationships between interrelated processes.
49. Explain the meaning of the term "critical area" of the process.
50. What are the real-world factors that might complicate the formulation of synchronization rules for interconnected processes?
51. What is a flow and what are flows in the OS?
52. What are the similarities and differences between process and flow?
53. At what level (processes or threads) is multiprogramming more effective and why?
54. Give the composition of the functional components of the OS and give a brief description of them.
55. List and explain the functions of the process control subsystem.
56. What is a process address space?
57. What is process context?
58. Explain the functions of file management and UVS subsystems and their relationship.

59. List and explain the functions of the memory management subsystem.
60. Describe the data protection and administration tools in the OS.
61. Who uses the API, a graphical user interface (GUI), and why?
62. What are the roles and capabilities of GUIs in modern operating systems?
63. Describe the concepts of building a GUI.
64. Describe the basic principles of building a GUI.
65. Describe examples of techniques and additional rules for building a GUI.
66. Describe the most significant events in the evolution of GUIs of popular operating systems and their families.
67. Show the need, role, and classes of interrupts.
68. What are the similarities and differences between interrupt handling and routine calling?
69. What is the difference between vector and pullable hardware interrupts?
70. What is an interrupt vector?
71. How is interrupt priority maintained?
72. What is the essence of interrupt masking?
73. What is the sequence of actions of the hardware and software to handle the interrupt?
74. How are software interrupts different from others?
75. What is virtualization and what does it do?
76. Give examples of virtualization.
77. How does virtualization manifest itself in the case of printer spooling?
78. What is the essence of virtual memory formation?
79. What is a virtual machine used for and in what ways?
80. What is the Resource Allocation Discipline (CRD) and what does it consist of?
81. What are the disciplines of queue formation and what factors affect them?
82. Draw a diagram of a circular cyclic algorithm and explain its features.
83. Compare different single-order CRRs.
84. Draw a diagram of the priority multi-sequence CRR and explain its features.
85. How will the behavior of the priority multi-sequence CRR change when switching to working with dynamic priorities?
86. What is the difference between absolute and relative priority service in CRR, and which service is easier to organize?
87. Compare different multi-sequence CRRs.
88. Draw a diagram of a priority-free multi-cycle CRR and explain its features.
89. Describe the options for organizing dynamic priority cyclic (carousel) CRR schemes.
90. What factors in real conditions complicate the solution of the problem of resource allocation?
91. Name the factors that complicate the allocation of resources using the example of OP and others.
92. What is the essence of the phenomenon of fragmentation of OC?
93. Give a detailed example of the occurrence of fragmentation.
94. What levels of user interaction with the computer are known, and which of them do you need special languages?
95. Describe the capabilities of OS command languages (with examples).
96. Give examples of command language operators for managing batch files.
97. Give examples of command language statements for configuring a PC environment.
98. Describe the main security tasks in the OS.
99. Characterize internal and external security threats in the OS.
100. Describe the objects and subjects of protection.
101. Describe the methods of user identification and authentication.
102. Describe the main security mechanisms (protection domains, access matrix, access control lists).
103. Describe the discretionary and mandatory access control models.
104. Describe the Bell-La Padula multi-level defense model.
105. Describe the Beebe's multi-level defense model in comparison with the Bell-La Padula model.
106. Describe the hidden channels of information transmission.
107. Describe the different security classes of information systems.
108. Describe the features of data protection in MS Windows and NTFS.

109. Describe the features of data protection in UNIX.
110. Describe encryption with a secret key.
111. Describe public-key encryption.
112. Describe the hybrid encryption scheme.
113. Describe digital signatures and certificates.
114. Describe the different types of insider attacks.
115. Give a classification and characteristics of malware.
116. Describe the different types of viruses.
117. Describe the different types of rootkits.
118. Describe the main methods of antivirus protection.
119. Describe the different types of firewalls and their features.
120. Describe the different types of antivirus programs and their features.

8.9. Test No 2 (written answer)

Type of assessment tool: test work

Type of control: milestone

List of controlled topics (modules): Module 2. Architecture and mechanisms of operating systems.

Form of study: full-time

Questions for the test No 2:

1. What is meant by the architecture, structure and composition of the OS?
2. What is the typical composition of fixed assets?
3. Introduce the classic architecture of the OS (based on the kernel), explain the composition and functions of the kernel and auxiliary modules of the OS.
4. What OS privilege modes should the computer hardware support?
5. How are applications subordinated to the operating system?
6. What is the impact of maintaining a multi-level privilege hierarchy?
7. What is the CPU switching latency typical for a classic OS architecture?
8. Draw a multi-layered structure of the OS and explain its main features.
9. List the features of layers and interlayer interfaces of the multilayer structure of the OS.
10. What are the advantages of a multi-layer OS structure?
11. List typical kernel layers and describe their functions.
12. Describe the features of the OS kernel resource manager layer.
13. What methods of layer interaction are used in classical architecture?
14. What does the increase/decrease in the number of OS kernel layers lead to?
15. List and describe typical OS hardware support tools.
16. Why can't the same OS be installed without changes on computers with a different type of processor or a way of organizing all the hardware?
17. Describe the features of the construction of machine-dependent components and the portability of the OS.
18. What is a "microkernel" and what modules are included in it?
19. What are "OS servers" in a microkernel-based architecture, in what mode do they work?

20. List and explain the features of the microkernel-based OS architecture.
21. Explain the mechanism for accessing OS functions in the form of microkernel-based architecture servers.
22. Describe the advantages and disadvantages of the microkernel-based OS architecture.
23. What is the CPU switching latency for a microkernel-based OS architecture?
24. What is the essence of compatibility of different operating systems and the features of its different types?
25. Why are libraries translated in application software environments?
26. Describe an option for implementing multiple application software environments based on system call translators.
27. Describe an option for implementing multiple application software environments based on support for multiple peer APIs.
28. Describe an option for implementing multiple application software environments based on the microkernel concept.
29. What does the presence of multiple application software environments in the OS give?
30. What actions does the OS perform when the process is generated?
31. What is a "process descriptor"?
32. What is a "flow descriptor"?
33. Explain the essence and main types of flow planning.
34. What is the difference between dynamic flow scheduling and static scheduling?
35. Explain the essence and procedure for dispatching flows.
36. Describe the composition of the flow context and the role of its hierarchical organization.
37. How are thread queues organized and reordered?
38. What is the difference between displacing and non-displacing planning algorithms?
39. Explain the features of planning algorithms based on quantization.
40. What is "flow priority" and what are its types?
41. How do dynamic thread priorities differ from static thread priorities?
42. Describe the prioritization scheme in Windows NT.
43. Describe the mixed scheduling algorithm in Windows NT.
44. Describe the mixed scheduling algorithm in UNIX System V Release 4.
45. Describe the mixed planning algorithm in OS/2.
46. Describe the scheme for changing the priorities of streams and the magnitude of quanta during planning in OS/2.
47. List the events that require CPU time reallocation and explain what the OS scheduler does in each case.
48. Describe the moments of redevelopment in the environment of the OS RV.
49. What additional organizational difficulties do interruptions create for the OS?
50. How are dispatching and interrupt prioritization in the OS carried out?
51. How does the interrupt manager work?
52. How do interrupt handler priorities relate to thread priorities?

53. How is interrupt dispatch consistent with thread dispatch?
54. How is the dispatching of system calls organized?
55. Describe the system call organization scheme with the System Call Manager.
56. Describe the features and differences in the organization of synchronous and asynchronous system calls.
57. What are the goals of the interaction of processes and flows?
58. Explain the essence and necessity of synchronizing processes and flows.
59. When do thread races occur?
60. What is a "critical section", "critical data", "mutual exclusion" of threads?
61. Explain the use of blocking variables.
62. Explain the use of semaphores.
63. Describe an example of using semaphores when working with a write/read buffer pool.
64. Explain the essence of mutual blockages (dead ends).
65. What is the difference between dead ends and queues?
66. Describe ideas and tools for identifying and resolving dead ends.
67. Explain the complexity of synchronizing the flows of different processes.
68. What methods does the OS use to separate synchronizing objects?
69. What are the usual OS objects that can be used as synchronizing ones, and what events put them in the signaling state?
70. What is the essence of the signal state of the synchronizing object of the fixed asset?
71. Give examples of signal states for the following synchronization objects: thread, process, file.
72. What is a mutex and an event object?
73. Explain the role of signals as synchronizing objects.
74. How do virtual addresses of commands and data differ from physical ones?
75. What is a virtual process address space and what parts is it divided into?
76. What methods of structuring the virtual address space of the process are used?
77. What is the "image of the process"?
78. Which areas of virtual memory are not preempted?
79. How is the use of different types of EP sections related to the phenomenon of fragmentation?
80. What classes are the algorithms for the distribution of EP divided into and which of them make up each class?
81. What tasks are solved during the virtualization of SF?
82. Name and explain the main approaches to virtualization of OC.
83. Explain the essence of swapping.
84. List the advantages and disadvantages of swapping.
85. What is a page table and what is it used for?
86. What information does the page handle include?
87. How is the paged distribution of EPs performed?
88. How is a virtual address presented in a page organization?

89. Describe known page replacement strategies.

90. How are sections supported during the paging distribution of EPs?

91. What disadvantages of page distribution does segment distribution eliminate?

92. What is the difference between the segment distribution of the OT and the page distribution?

93. How is the segmental distribution of the OP performed?

94. How is a virtual address presented in a segment organization?

95. What is a segment table and what is it used for?

96. What are the disadvantages and advantages of the segmental distribution of EP?

97. Explain the essence of the segment-page organization of the EP.

98. How does the transformation of a virtual address into a physical one take place during the segment-page organization of the SE?

99. How is the modified page mechanism used in the segment-page organization of the EP?

100. Give a brief comparison of different algorithms for the distribution of EP.

101. List and briefly explain the tasks of the OS for managing UVS and files.

102. Describe the need for and organization of parallel operation of the hydrocarbon and the processor, coordination of exchange and caching speeds.

103. Describe the need for and organization of the separation of the UVS and data between the processes, the logical interface between the UVS and the rest of the OS.

104. Describe the need for and organization of support for a wide range of drivers, dynamic driver loading and unloading.

105. Describe the need for and organization of support for multiple file systems, synchronous and asynchronous I/O operations.

106. Present and describe the generalized structure of the I/O subsystem.

107. Describe the organization and features of the I/O manager.

108. Describe the organization and features of layered drivers.

109. Describe the purpose and functions of the classic driver.

110. Describe the composition and purpose of the FS, as well as the tasks of the FS of different classes of fixed assets.

111. Describe the different file types that support file systems.

112. Describe file names, hierarchical structure and mounting of the file system, logical organization of the file.

113. Describe the organization of disks, their sectors, blocks, and clusters, disk formatting procedures, partitions, and their properties.

114. Describe the physical organization (location) and addressing of the file.

115. What is the difference between a linked list of indexes and a linked list of clusters?

Criteria for evaluating the test work:

− for answering each question of the option, the student can receive: 2 points - for a complete answer, 1 - for an incomplete answer, 0 - for an insufficient, incorrect or missing answer;

− the assessment of the test is determined as the sum of the points scored. The maximum amount of points for 5 questions in the option is 10 points.

8.10. Case Studies NoNo 1–6: A General Introduction to Cases

Before starting the first case, students receive the following information, which plays the role of a general introduction, explaining the tasks and the order of completing cases.

You are tasked with developing a structure and software for a complex specialized device, including a processor, memory, file storage, a set of inputs and outputs. Your management has decided to develop the device "from scratch" in order to achieve optimal quality indicators. At the same time, the concept of "quality" is not fully defined, but it is assumed that it includes reliability, performance, economy and, if possible, everything good in general.

The development of each of the subsystems of the device OS is considered as a separate case, the optimal solution of which depends on the characteristics of the subsystem given to you.

The student's task in each case is as follows:

1. Seek help from at least three AI systems of their choice, with the most accurate query possible.
2. Critically consider AI responses in terms of their feasibility and adequacy of the situation.
3. Refine AI queries as needed.
4. Make a decision on the case (taking into account the recommendations of the AI, if they are recognized as useful by the student).
5. Compile a progress report, including questions asked by the AI, AI responses (possibly abbreviated), and decisions made with justifications.

8.11. Cases NoNo 1–6 (completion, registration of results, interview, assessment)

Type of assessment tool: case protection.

Type of control: current.

Form of study: full-time.

List of controlled topics:

1. Selection of dispatching algorithms.

Example of initial data:

- The system can serve up to 10 real-time processes.
- The minimum input interval is 100 ms for each process.
- The system response time to input signals should not exceed 50 ms.
- The user can conduct a dialogue with the system, while the system reaction time to user actions should not exceed 300 ms.

2. Allocate memory to processes.

Example of initial data:

- The system's memory consists of 1 GB of RAM and a 10 GB solid-state drive.
- Memory is organized according to the page-based principle, calls to virtual addresses are translated into calls to physical addresses in real time using hardware.
- If there is not enough physical memory, the system can use disk memory to temporarily unload pages.
- The selection of pages for unloading is carried out according to the clock algorithm (second-chance algorithm with circular memory page viewing).
- The system uses low-load intervals to increase the number of free pages in the background.

3. Data protection.

Example of initial data:

- The system is multi-user. User authorization is performed on the basis of logins and passwords.

- Password guessing time should be unacceptably long.
- Two different processes do not have access to each other's programs and data by default.
- Simultaneous access of two or more processes to the same system object (for example, a file) is possible if this does not contradict the access modes of processes to the object specified for each process.
- Such simultaneous access is allowed, in particular, if all processes have read-only access to the object or if only one process has access rights.

4. Elimination of the effect of racing (mutual exclusion).

Example of initial data:
- In programs of two or more processes that want to access a system object, there are areas in which the simultaneous presence of processes can lead to data integrity.
- A process that is not in a similar part of the program should not interfere with the work of other processes.
- Processes are independent and equitable.

5. Organization of process interaction.

Example of initial data:
- All processes do not "see" each other by default.
- The two processes have the ability to synchronize their actions and/or exchange data by mutual agreement.
- Process synchronization means that one of the processes expects an event, the occurrence of which is ensured by another process.
- Data exchange can be unidirectional (one process only sends data, and the other only receives it) or bidirectional (each process can send data that is received by another process).
- Data exchange should be organized in such a way as not to create a problem of racing (data integrity violation).

6. File system structure.

Example of initial data:
- Program and data files are written to the device once and are not modified in the future.
- Each file is up to 100 MB in size.
- The directory structure is hierarchical, with no rigid or symbolic links.
- At least 1 TB of total file storage.
- Access to data is arbitrary.
- The probability of error when reading data or commands should be minimal.

Methodological recommendations for the implementation of cases.

The work is carried out in the classroom equipped in accordance with paragraph 7 of the Work Program of the discipline. The basis for admission to work is knowledge of theoretical material and methodological instructions that the student must demonstrate at the beginning of the lesson.

The process of work is documented using a text editor, the information obtained serves as the basis for the formation of a report.

The defense of the case report is accompanied by a demonstration of the results obtained, theoretical knowledge and answers to additional questions from the teacher on the topic of the lesson.

In the process of preparing and performing the work, the student is guided by the educational and methodological literature specified in paragraph 6 of the Work Program of the discipline.

Criteria for evaluating cases.

For each case, the student receives:

– 6 or 7 points, if he/she has completed all the tasks provided for in the work in a timely manner, prepared a report in accordance with the requirements of the teacher and, in the process of defense, demonstrated the completeness of theoretical knowledge in the amount of the content of the academic discipline related to the case; was able to answer additional questions related not only to the process of performing work, but also to the understanding of the actions performed and the tasks solved; there are no comments on the performance of work and the preparation of the report;

– 4 points if he/she has completed all the tasks provided for in the work, prepared a report in accordance with the requirements of the teacher and in the process of defense demonstrated the presence of sufficient theoretical knowledge in the scope of the content of the academic discipline related to the case; was able to answer questions related to the process of performing work; there are minor comments on the performance of work and the preparation of the report;

– 3 points if he/she has basically completed the tasks provided for in the work, prepared a report in accordance with the requirements of the teacher and demonstrated the presence of theoretical knowledge related to the work in the process of defense; there are significant comments on the performance of work, the preparation of the report or answers to questions;

– 2 points, if he/she has completed more than half of the tasks set in the work, is able to answer questions related to the theoretical component in the volume of the content of the academic discipline related to the case;

– 1 point, if the requirements provided for in case of obtaining 2 points are not met;

– 0 points if the work was not done and was not submitted.

8.12. Exam Questions and Tickets for Semester 3

Control period: 3rd semester

Type of assessment tool (subject of control): exam

List of controlled topics (modules): Fundamentals of operating systems. Architecture and

Operating system mechanisms

Form of the exam: oral questioning.

Evaluation criteria:

− the exam ticket contains 4 questions;

− For answering each question of the ticket, the student can receive up to 10 points:

− 9-10 points with excellent knowledge of the topic of the question and answers to all additional questions;

− 8 points with good knowledge of the topic of the question and answers to most of the additional questions;

− 6-7 points with satisfactory knowledge of the topic of the question and answers to certain additional questions;

− 0-5 points with an unsatisfactory level of knowledge and answers;

− after the end of the answers to the questions of the examination ticket, the points scored are summed up, the exam is considered passed with the total of 22-40 points.

List of exam questions:

1. Factors influencing the development of OS.
2. The concept and definitions of fixed assets.
3. Units of computing work in an OS environment. Criteria for the effectiveness of the Armed Forces, emphasizing the role of the environment.
4. The operating modes of computers provided by the OS.

5. Multiprogramming. Features, advantages and disadvantages of the multiprogram batch mode.
6. Example diagrams of 3 programs and CPU load for multiprogram batch mode.
7. Features, advantages and disadvantages of the RV mode.
8. Features, advantages and disadvantages of the RFD regime.
9. An example of diagrams of the operation of 3 programs and the CPU load for the RDV mode.
10. Features of multitasking mode and options for its implementation.
11. OS functions.
12. Classification of fixed assets with examples of known systems.
13. Principles of OS construction.
14. Requirements for fixed assets.
15. The evolution of popular operating systems and their families.
16. Trends in the development of fixed assets at the present stage.
17. Characteristics and definitions of the resource.
18. Concept and definitions of the process.
19. Process states, existence graph, existence interval, process trace.
20. Classification of processes.
21. Relations between interrelated processes.
22. Factors that complicate the formulation of synchronization rules for related processes.
23. Flow: concept, features, role in the OS environment.
24. Composition and brief characteristics of the functional components of the OS.
25. Functions of the process control subsystem.
26. The address space of the process, the context of the process.
27. Functions of the memory management subsystem.
28. Functions of file management and UVC subsystems, their interrelation.
29. Data protection and administration tools in the OS.
30. The role and capabilities of API, the concept of GUI.
31. Principles of construction of the Chief Project Engineer.
32. Techniques and additional rules for constructing the Chief Project Engineer.
33. Evolution of GUIs of popular operating systems and their families.
34. Necessity, role, and classes of interrupts.
35. Vector hardware interrupt mechanism.
36. The mechanism of interrogated hardware interrupts.
37. Interrupt prioritization and masking.
38. The sequence of actions of hardware and software to handle an interrupt.
39. Software interrupts.
40. Virtualization. Examples of virtualization.
41. Virtual memory and virtual machine.
42. Discipline of resource distribution, its components.

43. Single-order CRR.
44. Multi-order CRR.
45. Factors that complicate the solution of the problem of resource allocation in real conditions, including the example of OP.
46. The phenomenon of fragmentation of OP, an example and methods of struggle.
47. Options for organizing dynamic priority cyclic (carousel) CRR schemes.
48. Levels of user interaction with the computer. Features of OS command languages (with examples).
49. Security tasks in the OS. Internal and external security threats.
50. Objects and subjects of protection, methods of user identification and authentication.
51. Basic protection mechanisms.
52. Access control and multi-layered security models.
53. Hidden channels of information transmission, security classes of information systems.
54. Features of data protection in MS Windows and UNIX.
55. Encryption methods, hybrid scheme.
56. Digital signatures and certificates.
57. Types of insider attacks.
58. Classification and characteristics of malware, viruses and rootkits.
59. Methods of anti-virus protection.
60. Firewalls and antivirus programs.
61. Classic OS architecture (based on the kernel), composition and functions of the kernel and auxiliary OS modules.
62. OS privilege modes and tools. Multi-level hierarchy of privileges.
63. Multi-layer OS structure, features of layers and interlayer interfaces.
64. Typical OS kernel layers and their functions.
65. Features of the OS kernel resource manager layer.
66. Typical OS hardware support tools.
67. Features of the construction of machine-dependent components and OS portability.
68. Microkernel: concept and composition. OS servers.
69. Microkernel-based OS architecture. Mechanism for accessing the functions of the fixed asset.
70. New kernel and OS architecture options.
71. The essence and types of compatibility of various operating systems, library translation.
72. A variant of implementing multiple application software environments based on system call translators.
73. An option for implementing multiple application software environments based on support for multiple peer APIs.
74. A variant of implementing multiple application software environments based on the microkernel concept.
75. Hypervisors.
76. Virtualization technologies.

77. Types and implementations of hypervisors.
78. Actions of the OS when a process is generated.
79. Main types of flow scheduling.
80. Dispatch of threads. Thread queues. Composition and organization of the flow context.
81. Displacing and non-displacing planning algorithms.
82. Planning algorithms based on quantization.
83. Prioritization scheme in Windows NT.
84. Mixed scheduling algorithm in Windows NT.
85. Planning algorithm in UNIX System V Release 4.
86. Mixed planning algorithm in OS/2.
87. Scheme of changing the priorities of flows and the value of quanta during planning in OS/2.
88. Events that require CPU time remapping and OS scheduler actions in each case.
89. Moments of redevelopment in the environment of the OS RV.
90. Dispatching and accounting for interrupt priorities in the OS. Interrupt Manager.
91. System call dispatching.
92. Diagram of the organization of system calls with the system call manager.
93. Features and differences in the organization of synchronous and asynchronous system calls.
94. Interaction goals, synchronization of processes and flows. Races, critical section, mutual exclusion of flows.
95. Blocking variables and semaphores. Examples.
96. An example of using semaphores when working with a read write buffer pool.
97. Dead ends, ideas and means of their identification and elimination.
98. Types of synchronizing OS objects, their signal state, examples.
99. A mutex, an event object. Signals.
100. Types of command and data addresses. UARP and the image of the process.
101. Ways of structuring the UARP process.
102. Classes of EP distribution algorithms, composition of classes.
103. Swapping, its advantages and disadvantages.
104. Paged distribution of EP.
105. Known page replacement strategies.
106. Section support for paged OP distribution.
107. Segment distribution of the OP.
108. Segment-page distribution of EP.
109. Shared memory segments.
110. OS tasks for managing UVS and files.
111. Features of the most important tasks of the I/O subsystem.
112. Generalized structure of the input-output subsystem.
113. Organization and features of the I/O manager.

114. Organization and features of multi-level drivers.
115. The purpose and functions of the classic driver.
116. Composition and purpose of the FS, tasks of the FS of different classes of fixed assets.
117. Different file types that FS supports.
118. File names, hierarchical structure and mounting of the file system, logical organization of the file.
119. Organization of disks, their sectors, blocks and clusters, procedures for formatting disks, partitions and their properties.
120. Physical organization (location) and file addressing.

Sample exam ticket:

MINISTRY OF EDUCATION AND SCIENCE OF THE RUSSIAN FEDERATION
Federal State Autonomous Educational
Institution of higher education
"SOUTHERN FEDERAL UNIVERSITY"

EXAM TICKET No 1

in the discipline OPERATING SYSTEMS

1. The concept and definitions of fixed assets.
2. Mechanism of vector hardware interrupts.
3. Swapping, its advantages and disadvantages.
4. Physical organization (placement) and addressing of the file.

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________________________ S. N. Drozdov

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